

SECTION 1: DOUBLE PARALLEL FILLET WELD

System Parameters: Plate Width $b = 100\text{ mm}$, Thickness $s = 10\text{ mm}$, Static Load $P = 80\text{ kN} = 80 \times 10^3\text{ N}$, Permissible Shear Stress $\tau = 55\text{ MPa} = 55\text{N/mm}^2$.

Design Protocol: Calculate effective weld length l for double parallel fillet weld under static load and total weld run length L adding 12.5 mm start/stop allowance.

1. Effective Length of Weld (l)

Double parallel fillet weld load capacity

$$\begin{aligned} P &= 1.414 \cdot s \cdot l \cdot \tau \\ 80 \times 10^3 &= 1.414 \times 10 \times l \times 55 \\ l &= \frac{80 \times 10^3}{778} \\ &= 102.8\text{ mm} \\ \Rightarrow l &= 103\text{ mm} \end{aligned}$$

2. Total Length of Weld Run (L)

Adding 12.5 mm allowance for starting and stopping

$$\begin{aligned} L &= l + 12.5\text{ mm} \\ &= 103 + 12.5 \\ &= 115.5\text{ mm} \end{aligned}$$

3. Design Output Summary

Sized Double Parallel Fillet Weld

Total Length of Weld Run $L = 115.5\text{ mm}$

SECTION 2: CIRCULAR FILLET WELD UNDER TORSION

System Parameters: Shaft Diameter $d = 50 \text{ mm}$, Weld Size $s = 10 \text{ mm}$, Allowable Shear Stress $\tau_{\max} = 80 \text{ MPa} = 80\text{N/mm}^2$.

Design Protocol: Determine maximum torque capacity T of circular fillet weld joining solid shaft to flat plate.

1. Maximum Torque Capacity (T)

Torsional shear stress in circular fillet weld

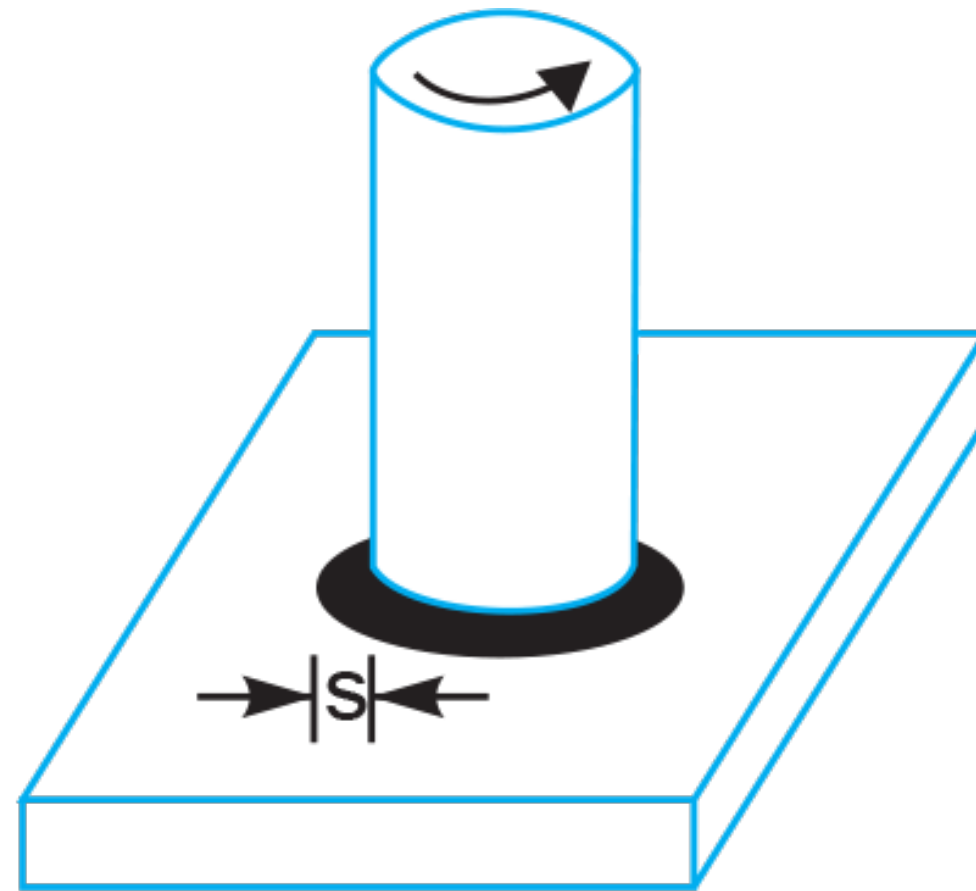
$$\begin{aligned}\tau_{\max} &= \frac{2.83T}{\pi \cdot s \cdot d^2} \\ 80 &= \frac{2.83T}{\pi \times 10 \times (50)^2} \\ 80 &= \frac{2.83T}{78550} \\ T &= \frac{80 \times 78550}{2.83} \\ &= 2.22 \times 10^6 \text{ N} \cdot \text{mm} \\ &= \mathbf{2.22 \text{ kN} \cdot \text{m}}\end{aligned}$$

2. Design Output Summary

Circular Fillet Weld Torque Capacity

Maximum Torque Capacity $T = 2.22 \text{ kN.m}$

SECTION 2: CIRCULAR FILLET WELD UNDER TORSION — MECHANICAL DIAGRAM

**Fig. 10.12****Figure 10.12: Solid shaft welded to plate by circular fillet weld under torsion**

SECTION 3: LONG FILLET WELD UNDER TORSION

System Parameters: Plate Length $l = 1 \text{ m} = 1000 \text{ mm}$, Thickness = 60 mm, Weld Size $s = 15 \text{ mm}$, Permissible Shear Stress $\tau_{\max} = 80 \text{ MPa} = 80\text{N/mm}^2$.

Design Protocol: Calculate maximum torque capacity T for long fillet weld connecting two plates at right angles.

1. Maximum Torque Capacity (T)

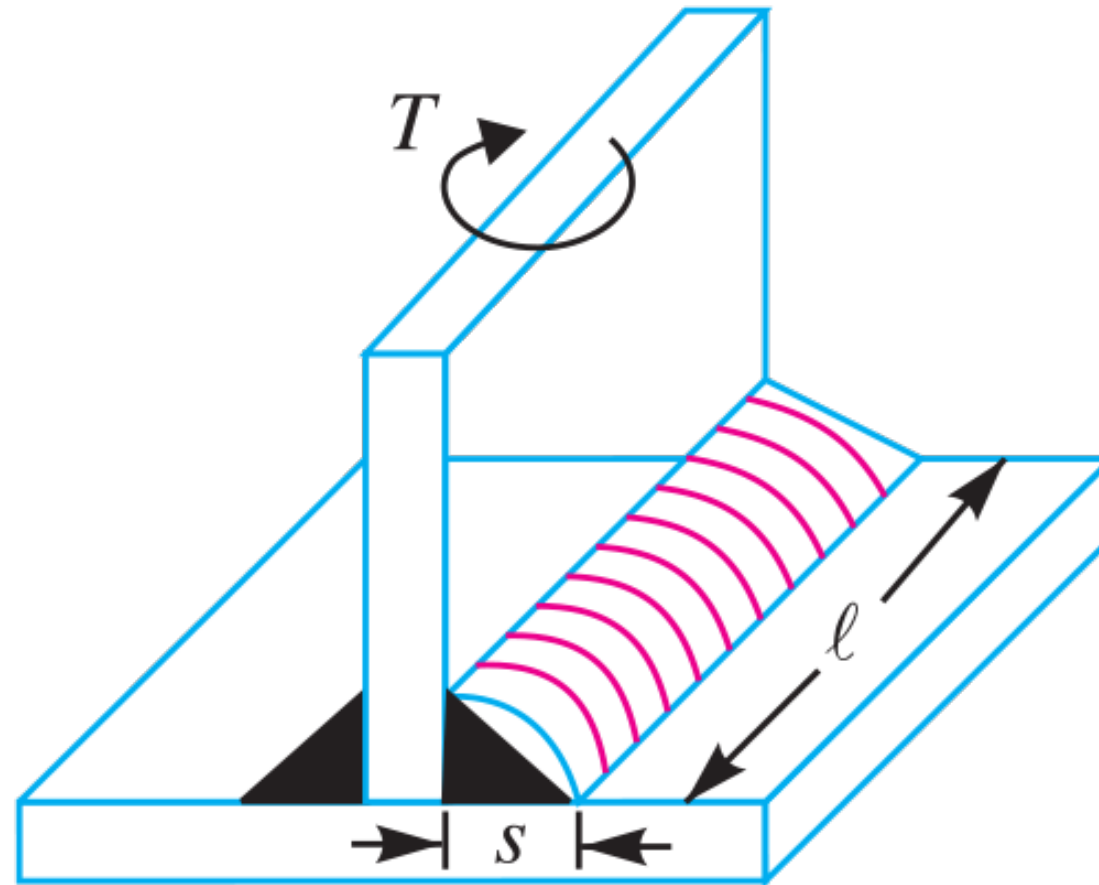
Torsional shear stress in long fillet weld

$$\begin{aligned}\tau_{\max} &= \frac{4.242T}{s \cdot l^2} \\ 80 &= \frac{4.242T}{15 \times (1000)^2} \\ 80 &= \frac{0.283T}{10^6} \\ T &= \frac{80 \times 10^6}{0.283} \\ &= 283 \times 10^6 \text{ N} \cdot \text{mm} \\ &= \mathbf{283 \text{ kN} \cdot \text{m}}\end{aligned}$$

2. Design Output Summary

Long Fillet Weld Torque Capacity

Maximum Torque Capacity $T = 283 \text{ kN.m}$

SECTION 3: LONG FILLET WELD UNDER TORSION — MECHANICAL DIAGRAM**Fig. 10.13****Figure 10.13: Long fillet weld subjected to torsion**

SECTION 4: PARALLEL FILLET WELD — STATIC & FATIGUE LOADING

System Parameters: Plate Width $b = 100$ mm, Weld Size $s = 12.5$ mm, Load $P = 50$ kN = 50×10^3 N, Allowable Shear Stress $\tau = 56$ MPa, Stress Concentration Factor $K_f = 2.7$.

Design Protocol: Calculate required weld run length under static loading and fatigue loading.

1. Length of Weld for Static Loading (l)

Load capacity for double parallel fillet weld

$$\begin{aligned} P &= 1.414 \cdot s \cdot l \cdot \tau \\ 50 \times 10^3 &= 1.414 \times 12.5 \times l \times 56 \\ l &= \frac{50 \times 10^3}{990} \\ &= 50.5 \text{ mm} \end{aligned}$$

2. Total Static Weld Length (L)

Adding 12.5 mm start/stop allowance

$$\begin{aligned} L &= l + 12.5 \text{ mm} \\ &= 50.5 + 12.5 \\ &= \mathbf{63 \text{ mm}} \end{aligned}$$

3. Permissible Shear Stress under Fatigue (τ_{fatigue})

Fatigue stress concentration factor $K_f = 2.7$

$$\begin{aligned}\tau_{\text{fatigue}} &= \frac{\tau}{K_f} \\ &= \frac{56}{2.7} \\ &= 20.74\text{N/mm}^2\end{aligned}$$

4. Total Fatigue Weld Length (L_{fatigue})

Effective length and total run length under fatigue loading

$$\begin{aligned}P &= 1.414 \cdot s \cdot l_{\text{fatigue}} \cdot \tau_{\text{fatigue}} \\ 50 \times 10^3 &= 1.414 \times 12.5 \times l_{\text{fatigue}} \times 20.74 \\ l_{\text{fatigue}} &= \frac{50 \times 10^3}{367} \\ &= 136.2 \text{ mm} \\ L_{\text{fatigue}} &= 136.2 + 12.5 \\ &= 148.7 \text{ mm}\end{aligned}$$

5. Design Output Summary

Parallel Fillet Weld Lengths

Static Length L = 63 mm, Fatigue Length L_fatigue = 148.7 mm

SECTION 5: COMBINATION WELD — STATIC & FATIGUE LOADING

System Parameters: Width $b = 75$ mm, Thickness $s = 12.5$ mm, Tensile Stress $\sigma_t = 70$ MPa, Shear Stress $\tau = 56$ MPa, $K_{f1} = 1.5$ (transverse), $K_{f2} = 2.7$ (parallel).

Design Protocol: Determine length of each parallel fillet weld for both static and fatigue loading.

1. Total Plate Load Capacity (P)

Maximum tensile load plate can carry

$$P = b \cdot s \cdot \sigma_t$$
$$= 75 \times 12.5 \times 70$$
$$= 65625 \text{ N}$$

2. Transverse Weld Effective Length (l_1)

Width minus start/stop allowance

$$l_1 = b - 12.5 \text{ mm}$$
$$= 75 - 12.5$$
$$= 62.5 \text{ mm}$$

3. Parallel Fillet Length for Static Loading (l_2)

Load sharing between transverse and parallel welds

$$P = P_1 + P_2$$

$$P_1 = 0.707s \cdot l_1 \cdot \sigma_t$$

$$= 0.707 \times 12.5 \times 62.5 \times 70 = 38664 \text{ N}$$

$$P_2 = 1.414s \cdot l_2 \cdot \tau$$

$$= 1.414 \times 12.5 \times l_2 \times 56 = 990l_2$$

$$65625 = 38664 + 990l_2$$

$$l_2 = \frac{65625 - 38664}{990}$$

$$= 27.2 \text{ mm}$$

$$\Rightarrow l_2 + 12.5 = \mathbf{40 \text{ mm}}$$

4. Parallel Fillet Length for Fatigue Loading
(*l*₂, fatigue)

Fatigue permissible stresses and load balancing

$$\begin{aligned}\sigma_{t, \text{ fatigue}} &= \frac{70}{1.5}, & \tau_{\text{fatigue}} &= \frac{56}{2.7} \\ \sigma_{t, \text{ fatigue}} &= 46.7\text{N/mm}^2 \\ \tau_{\text{fatigue}} &= 20.74\text{N/mm}^2 \\ P_1 &= 0.707(12.5)(62.5)(46.7) = \\ P_2 &= 1.414(12.5)l_2(20.74) = 366l_2 \\ 65625 &= 25795 + 366l_2 \\ l_2 &= \frac{65625 - 25795}{366} \\ &= 108.8 \text{ mm} \\ \Rightarrow l_2 + 12.5 &= \mathbf{121.3 \text{ mm}}\end{aligned}$$

5. Design Output Summary
Combination Weld Lengths

Static Parallel Weld *l*₂ = 40 mm, Fatigue Parallel Weld *l*₂ =
121.3 mm

SECTION 5: COMBINATION TRANSVERSE AND PARALLEL FILLET WELD — MECHANICAL DIAGRAM

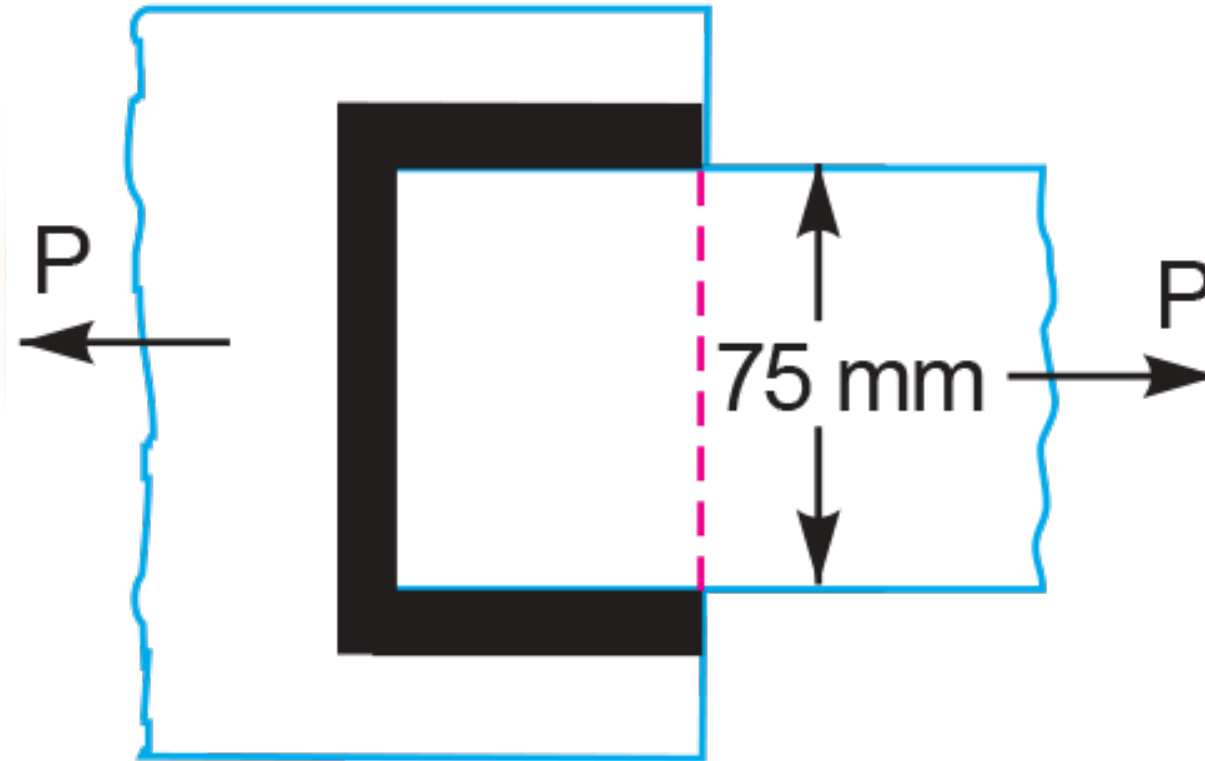


Fig. 10.15

Figure 10.15: Combination of single transverse and double parallel fillet weld

SECTION 6: WELD RUN LENGTH UNDER VARIABLE LOADING

System Parameters: Plate Width $b = 120$ mm, Thickness $s = 15$ mm, Static Tensile Stress $\sigma_t = 70$ MPa, Static Shear Stress $\tau = 56$ MPa, $K_{f1} = 1.5$, $K_{f2} = 2.7$.

Design Protocol: Determine weld run lengths for single transverse weld and double parallel fillet welds.

1. Single Transverse Weld Length (l_1)

Width minus start/stop allowance

$$\begin{aligned} l_1 &= b - 12.5 \text{ mm} \\ &= 120 - 12.5 \\ &= \mathbf{107.5 \text{ mm}} \end{aligned}$$

2. Double Parallel Fillet Weld Length (l_2)

Plate load capacity and fatigue load balance

$$\begin{aligned} P &= b \cdot s \cdot \sigma_t \\ &= 120 \times 15 \times 70 \\ &= 126 \times 10^3 \text{ N} \\ P_1 &= 0.707(15)(107.5)(46.7) = 53240 \text{ N} \\ P_2 &= 1.414(15)l_2(20.74) = 440l_2 \\ 126 \times 10^3 &= 53240 + 440l_2 \\ l_2 &= \frac{126000 - 53240}{440} \\ &= 165.4 \text{ mm} \\ \Rightarrow l_2 + 12.5 &= \mathbf{178 \text{ mm}} \end{aligned}$$

3. Design Output Summary

Dynamic Load Weld Run Lengths

Transverse Weld Length $l_1 = 107.5$ mm, Double Parallel
Fillet Length $l_2 = 178$ mm

SECTION 6: LONG FILLET WELD UNDER VARIABLE LOADING — MECHANICAL DIAGRAM

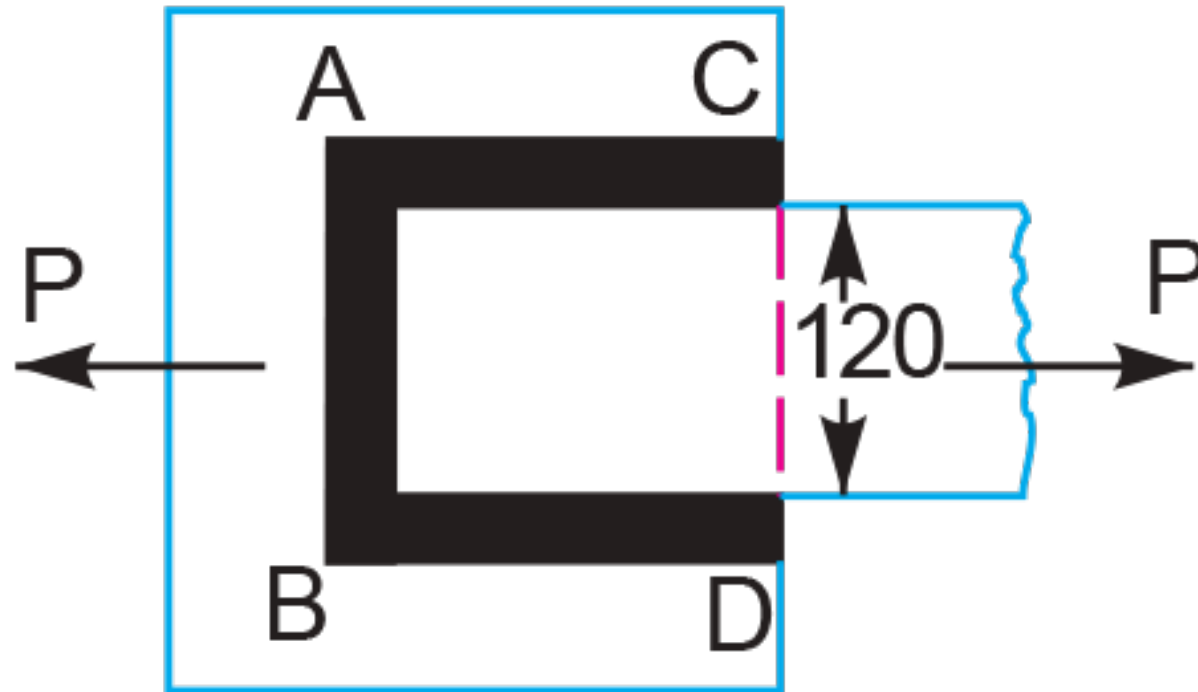


Fig. 10.16

Figure 10.16: Long fillet weld subjected to dynamic variable loading

SECTION 7: FILLET WELD 'T' SECTION SHEAR PLANE ANALYSIS

System Parameters: Leg size s , Weld length l .

Design Protocol: Locate maximum shear plane angle θ for parallel and transverse loading, and find ratio of limiting loads.

1. Shear Plane for Parallel Loading (θ)

Maximizing shear stress by differentiation

$$\tau = \frac{P(\cos \theta + \sin \theta)}{2s \cdot l}$$
$$\frac{d\tau}{d\theta} = \frac{P}{2s \cdot l}(-\sin \theta + \cos \theta) = 0$$
$$\sin \theta = \cos \theta$$
$$\Rightarrow \theta = 45^{\text{deg}}$$
$$P_{\text{parallel}} = 1.414 \cdot s \cdot l \cdot \tau_{\text{max}}$$

2. Shear Plane for Transverse Loading (θ)

Normal and shear force equilibrium resolution

$$\tau = \frac{P \sin \theta (\cos \theta + \sin \theta)}{2s \cdot l}$$

$$\frac{d\tau}{d\theta} = 0$$

$$\cos 2\theta + \sin 2\theta = 0$$

$$\tan 2\theta = -1$$

$$2\theta = 135^{\text{deg}}$$

$$\Rightarrow \theta = 67.5^{\text{deg}}$$

$$P_{\text{transverse}} = 1.65 \cdot s \cdot l \cdot \tau_{\text{max}}$$

3. Ratio of Limiting Loads

Ratio of parallel load capacity to transverse load capacity

$$\text{Ratio} = \frac{P_{\text{parallel}}}{P_{\text{transverse}}}$$

$$= \frac{1.414s \cdot l \cdot \tau_{\text{max}}}{1.65s \cdot l \cdot \tau_{\text{max}}}$$

$$= 0.857$$

4. Design Output Summary

T-Section Shear Planes

Parallel Plane theta = 45 deg, Transverse Plane theta = 67.5 deg, Load Ratio = 0.857

SECTION 7: FILLET WELD 'T' SECTION — MECHANICAL DIAGRAM

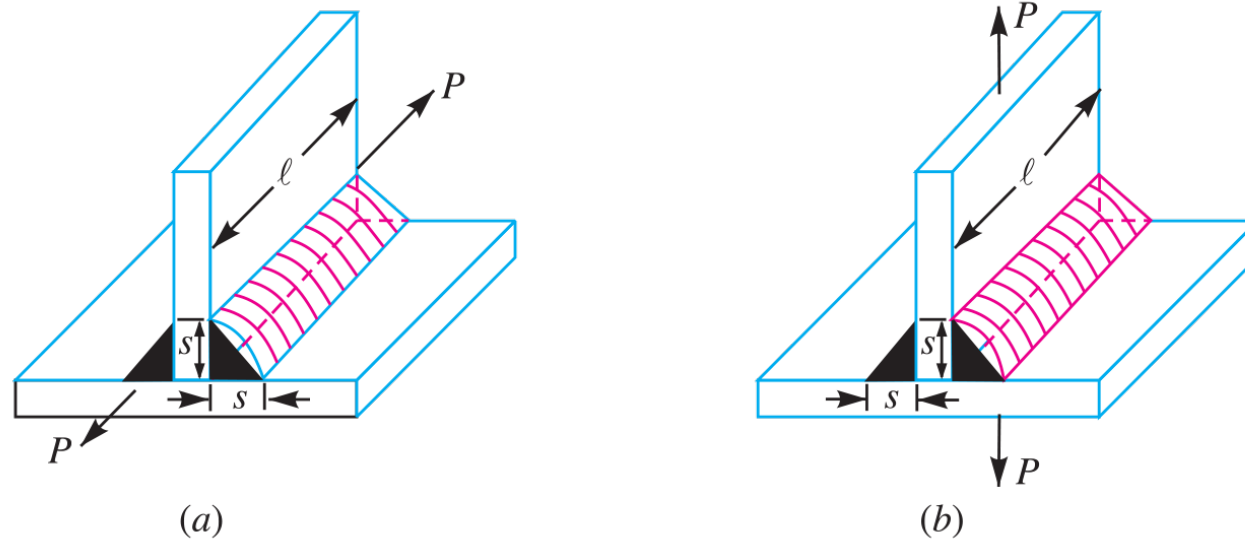


Fig. 10.17

Figure 10.17: Fillet welds fabricating a T-section under parallel and transverse loads

SECTION 7: PARALLEL LOAD SHEAR PLANE GEOMETRY — MECHANICAL DIAGRAM

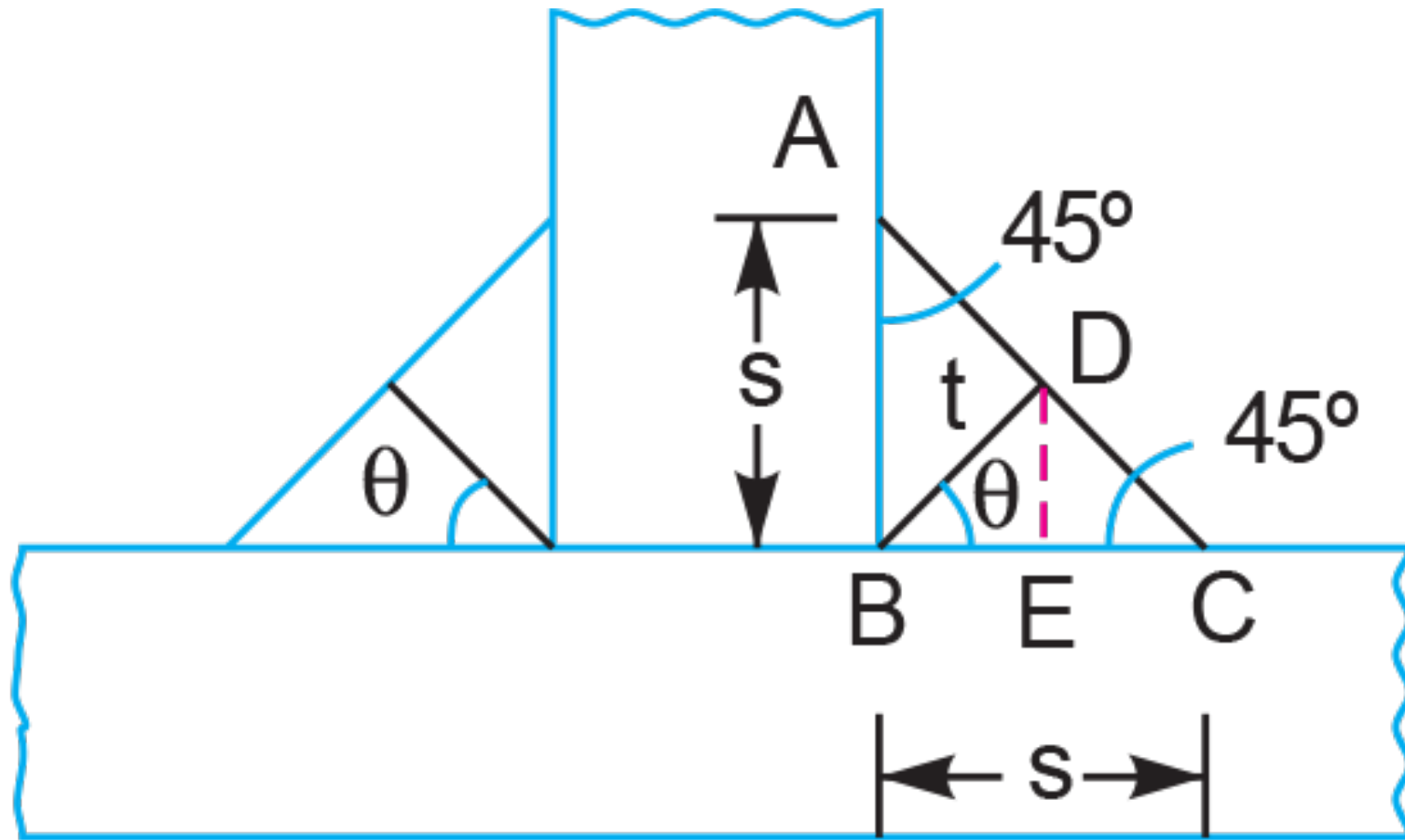
**Fig. 10.18**

Figure 10.18: Throat plane dimensions for load parallel to weld

SECTION 7: TRANSVERSE LOAD FORCE RESOLUTION — MECHANICAL DIAGRAM

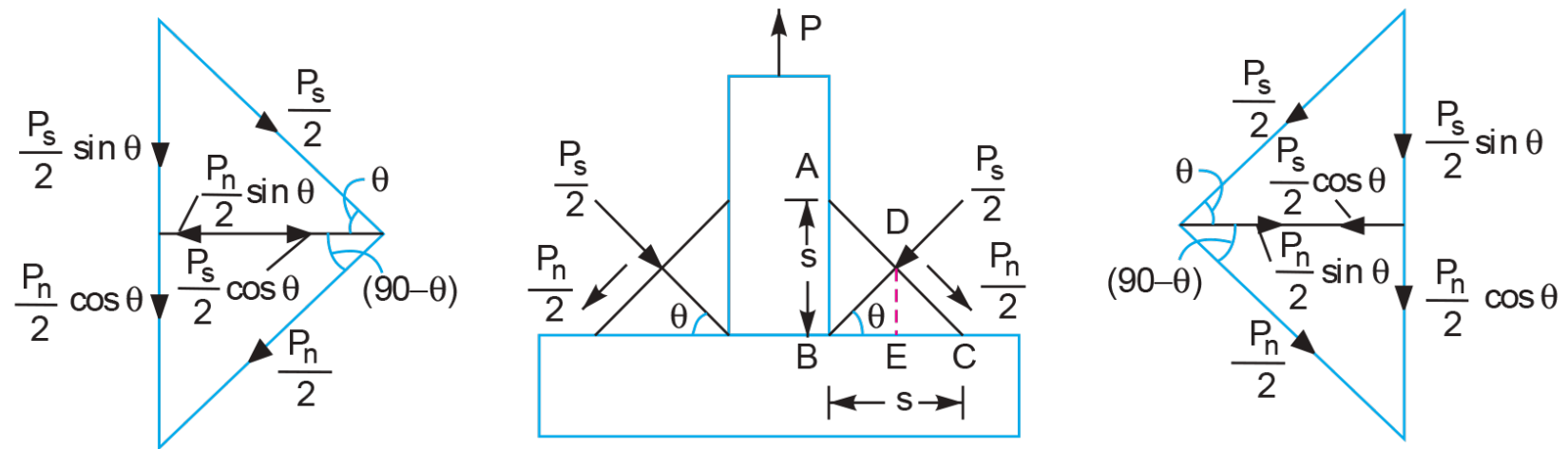


Fig. 10.19

Figure 10.19: Normal and shear force components for transverse load

SECTION 8: AXIALLY LOADED UNSYMMETRICAL SECTION ANGLE WELDING

System Parameters: Angle $200 \times 150 \times 10$ mm ($a + b = 200$ mm), Static Load $P = 200$ kN $= 200 \times 10^3$ N, Allowable Shear Stress $\tau = 75$ MPa, Weld Size $s = 10$ mm.

Design Protocol: Determine top (l_a) and bottom (l_b) weld lengths by balancing load about centroidal axis.

1. Total Required Weld Length (l)

Load capacity for single parallel fillet welds

$$\begin{aligned} P &= 0.707 \cdot s \cdot l \cdot \tau \\ 200 \times 10^3 &= 0.707 \times 10 \times l \times 75 \\ l &= \frac{200 \times 10^3}{530.25} \\ &= 377 \text{ mm} \end{aligned}$$

2. Position of Centroidal Axis (a, b)

Distance of gravity axis from bottom edge

$$\begin{aligned} b &= \frac{(200 - 10) \times 10 \times 95 + 150 \times 10 \times 5}{190 \times 10 + 150 \times 10} \\ &= \frac{188000}{3400} \\ &= 55.3 \text{ mm} \\ a &= 200 - 55.3 \end{aligned}$$

3. Top and Bottom Weld Lengths (l_a, l_b)

Moment equilibrium about centroidal line

$$\begin{aligned}
 l_a &= l \cdot \left(\frac{b}{a + b} \right) \\
 &= 377 \times \left(\frac{55.3}{200} \right) \\
 &= 104.2 \text{ mm} \\
 l_b &= 377 - 104.2 \\
 &= 272.8 \text{ mm}
 \end{aligned}$$

4. Design Output Summary

Unsymmetrical Angle Weld Lengths

Top Weld Length l_a = 104.2 mm, Bottom Weld Length l_b = 272.8 mm

SECTION 8: UNSYMMETRICAL ANGLE SECTION WELDING — MECHANICAL DIAGRAM

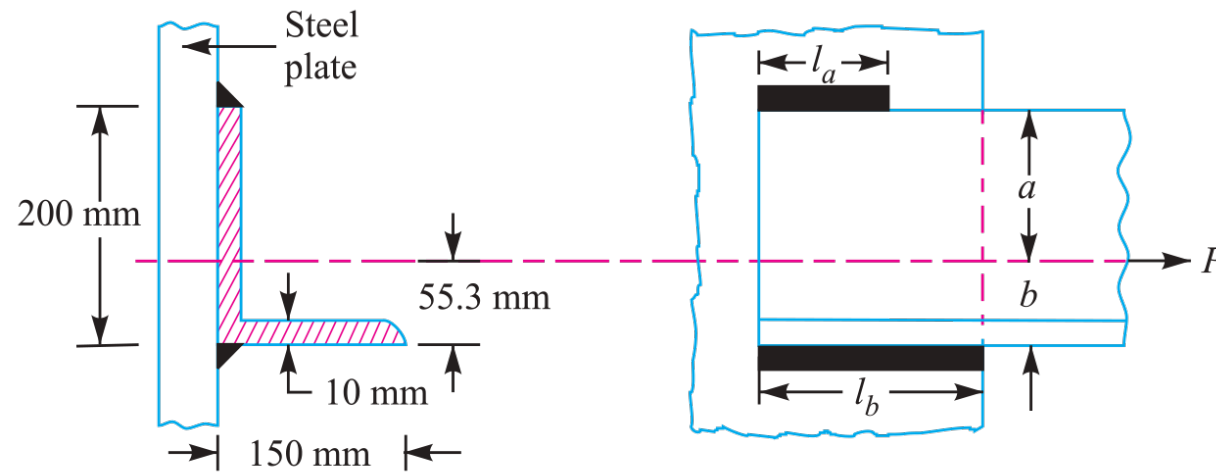


Fig. 10.21

Figure 10.21: Unsymmetrical angle section welded to plate with top and bottom welds

SECTION 9: ECCENTRICALLY LOADED OUT-OF-PLANE WELDED JOINT

System Parameters: Load $P = 2 \text{ kN} = 2000 \text{ N}$, Eccentricity $e = 120 \text{ mm}$, Weld Length $l = 40 \text{ mm}$, Allowable Max Shear Stress $\tau_{\text{max}} = 25 \text{ MPa}$.

Design Protocol: Determine weld size s under combined direct shear and bending.

1. Direct Shear Stress (τ)

Throat area of double fillet weld

$$\begin{aligned} \tau &= \frac{P}{A} = \frac{2000}{1.414s \cdot l} \\ &= \frac{2000}{1.414s \times 40} \\ &= \frac{35.4}{s} \text{ N/mm}^2 \end{aligned}$$

2. Bending Stress (σ_b)

Bending moment and throat section modulus

$$\begin{aligned} \sigma_b &= \frac{M}{Z} = \frac{P \cdot e}{\frac{s \cdot l^2}{4.242}} \\ &= \frac{2000 \times 120}{\frac{s \times (40)^2}{4.242}} \\ &= \frac{636.6}{s} \text{ N/mm}^2 \end{aligned}$$

3. Maximum Shear Stress & Weld Size (s)

Maximum shear stress theory

$$\tau_{\max} = \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau^2}$$

$$25 = \frac{1}{2} \sqrt{\left(\frac{636.6}{s}\right)^2 + 4\left(\frac{35.4}{s}\right)^2}$$

$$25 = \frac{320.3}{s}$$

$$s = \frac{320.3}{25}$$

$$= 12.8 \text{ mm}$$

4. Design Output Summary

Out-of-Plane Joint Weld Size

Required Weld Size $s = 12.8 \text{ mm}$

SECTION 9: ECCENTRICALLY LOADED OUT-OF-PLANE WELDED JOINT — MECHANICAL DIAGRAM

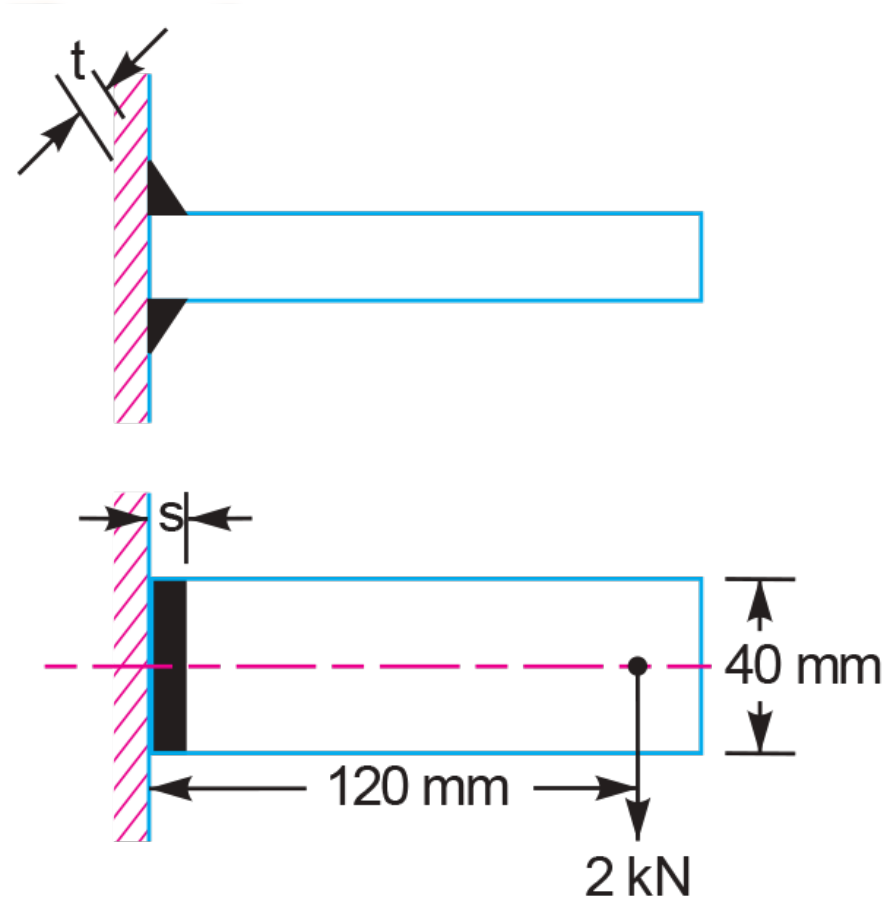


Fig. 10.24

Figure 10.24: Joint subjected to out-of-plane eccentric load

SECTION 10: CIRCULAR FILLET WELD UNDER ECCENTRIC BENDING LOAD

System Parameters: Shaft Diameter $D = 50$ mm, Weld Size $s = 15$ mm, Load $P = 10$ kN = 10000 N, Eccentricity $e = 200$ mm.

Design Protocol: Determine maximum normal and shear stresses in circular fillet weld.

1. Direct Shear Stress (τ)

Throat area for circular weld

$$\begin{aligned}\tau &= \frac{P}{0.707s \cdot \pi D} \\ &= \frac{10000}{0.707 \times 15 \times \pi \times 50} \\ &= 6\text{N/mm}^2 = 6 \text{ MPa}\end{aligned}$$

2. Bending Stress (σ_b)

Bending moment and circular section modulus

$$\begin{aligned}\sigma_b &= \frac{M}{Z} = \frac{P \cdot e}{\frac{\pi}{4}(0.707s)D^2} \\ &= \frac{10000 \times 200}{\frac{\pi}{4}(0.707 \times 15)(50)^2} \\ &= \frac{2 \times 10^6}{20825} \\ &= 96\text{N/mm}^2 = 96 \text{ MPa}\end{aligned}$$

3. Maximum Normal Stress ($\sigma_{t(max)}$)

Principal stress theory

$$\begin{aligned}\sigma_{t(max)} &= \frac{1}{2}\sigma_b + \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2} \\ &= 48 + \frac{1}{2}\sqrt{(96)^2 + 4(6)^2} \\ &= 48 + 48.4 \\ &= \mathbf{96.4\ MPa}\end{aligned}$$

4. Maximum Shear Stress (τ_{max})

Maximum shear stress theory

$$\begin{aligned}\tau_{max} &= \frac{1}{2}\sqrt{\sigma_b^2 + 4\tau^2} \\ &= \frac{1}{2}\sqrt{(96)^2 + 4(6)^2} \\ &= \mathbf{48.4\ MPa}\end{aligned}$$

5. Design Output Summary

Circular Weld Stresses

Max Normal Stress = 96.4 MPa, Max Shear Stress = 48.4 MPa

SECTION 10: CIRCULAR SHAFT WELDED TO PLATE UNDER BENDING — MECHANICAL DIAGRAM

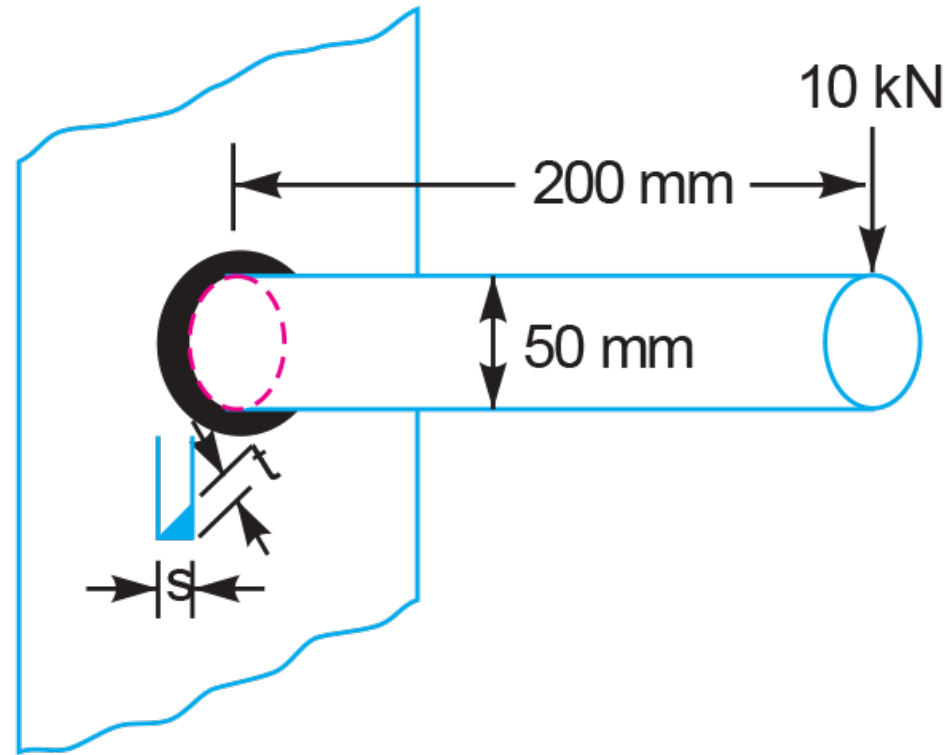


Fig. 10.25

Figure 10.25: Solid shaft welded to plate subjected to eccentric bending load

SECTION 11: RECTANGULAR CROSS-SECTION BAR WELDED TO SUPPORT

System Parameters: Load $P = 25 \text{ kN} = 25 \times 10^3 \text{ N}$, Eccentricity $e = 500 \text{ mm}$, $l = 100 \text{ mm}$, $b = 150 \text{ mm}$, Permissible Shear Stress $\tau_{\text{max}} = 75 \text{ MPa}$.

Design Protocol: Calculate weld size s for rectangular bar welded on 4 sides.

1. Direct Shear Stress (τ)

Throat area around all 4 sides

$$\begin{aligned} \tau &= \frac{P}{0.707s(2b + 2l)} \\ &= \frac{25 \times 10^3}{0.707s(2 \times 150 + 2 \times 100)} \\ &= \frac{70.72}{s} \text{N/mm}^2 \end{aligned}$$

2. Bending Stress (σ_b)

Bending moment and throat section modulus

$$\begin{aligned}\sigma_b &= \frac{M}{Z} = \frac{P \cdot e}{0.707s \left(\frac{b^2}{3} + b \cdot l \right)} \\ &= \frac{25 \times 10^3 \times 500}{0.707s \left(\frac{(150)^2}{3} + 150 \times 100 \right)} \\ &= \frac{12.5 \times 10^6}{15907.5s} \\ &= \frac{785.8}{s} \text{ N/mm}^2\end{aligned}$$

3. Maximum Shear Stress & Weld Size (s)

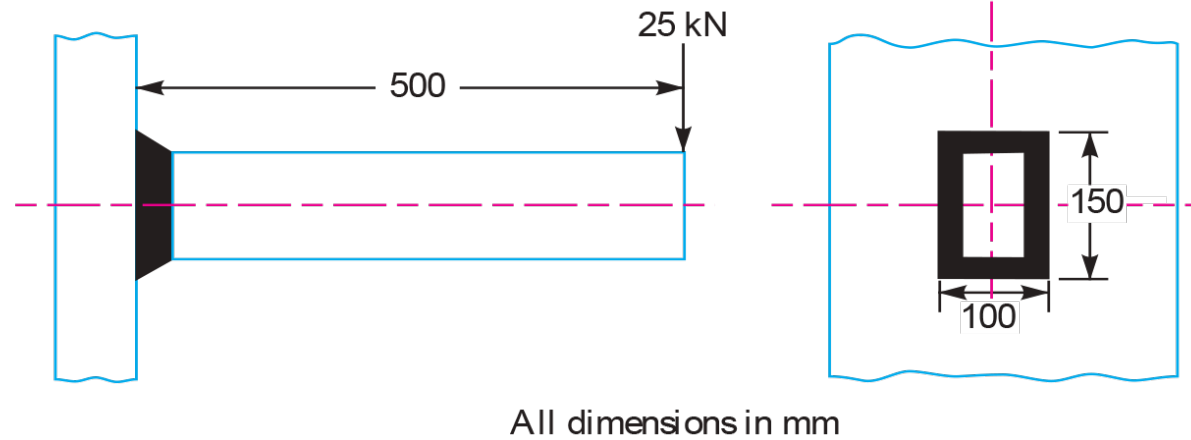
Maximum shear stress theory

$$\begin{aligned}\tau_{\max} &= \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau^2} \\ 75 &= \frac{1}{2} \sqrt{\left(\frac{785.8}{s} \right)^2 + 4 \left(\frac{70.72}{s} \right)^2} \\ 75 &= \frac{399.2}{s} \\ s &= \frac{399.2}{75} \\ &= \mathbf{5.32 \text{ mm}}\end{aligned}$$

4. Design Output Summary

Rectangular Bar Weld Size

Required Weld Size $s = 5.32 \text{ mm}$

SECTION 11: RECTANGULAR BAR WELDED TO SUPPORT — MECHANICAL DIAGRAM**Fig. 10.26****Figure 10.26: Rectangular cross-section bar welded on 4 sides subjected to eccentric load**

SECTION 12: ARM WELDED TO HOLLOW SHAFT UNDER BENDING AND TORSION

System Parameters: Force $P = 15 \text{ kN} = 15 \times 10^3 \text{ N}$, Shaft Diameter $d = 80 \text{ mm}$, Torque Arm 240 mm, Bending Arm 175 mm, Permissible Shear Stress $\tau_{\max} = 120 \text{ MPa}$.

Design Protocol: Determine weld size s for combined torsion and bending.

1. Torsional Shear Stress (τ)

Torque acting on shaft and torsional shear

$$\tau = \frac{2.83T}{\pi \cdot s \cdot d^2}$$

$$T = 15 \times 10^3 \times 240 = 3600 \times 10^3 \text{ N} \cdot \text{mm}$$

$$\tau = \frac{2.83 \times 3600 \times 10^3}{\pi \cdot s \times (80)^2}$$

$$= \frac{506.6}{s} \text{ N/mm}^2$$

2. Bending Stress (σ_b)

Bending moment and bending stress

$$\sigma_b = \frac{5.66M}{\pi \cdot s \cdot d^2}$$

$$M = 15 \times 10^3 \times (200 - 25) = 2625 \times 10^3 \text{ N} \cdot \text{mm}$$

$$\sigma_b = \frac{5.66 \times 2625 \times 10^3}{\pi \cdot s \times (80)^2}$$

$$= \frac{738.8}{s} \text{ N/mm}^2$$

3. Maximum Shear Stress & Weld Size (s)

Maximum shear stress theory

$$\tau_{\max} = \frac{1}{2} \sqrt{\sigma_b^2 + 4\tau^2}$$

$$120 = \frac{1}{2} \sqrt{\left(\frac{738.8}{s}\right)^2 + 4\left(\frac{506.6}{s}\right)^2}$$

$$120 = \frac{627}{s}$$

$$s = \frac{627}{120}$$

$$= \mathbf{5.2 \text{ mm}}$$

4. Design Output Summary

Arm to Shaft Weld Size

Required Weld Size $s = 5.2 \text{ mm}$

SECTION 12: ARM WELDED TO HOLLOW SHAFT — MECHANICAL DIAGRAM

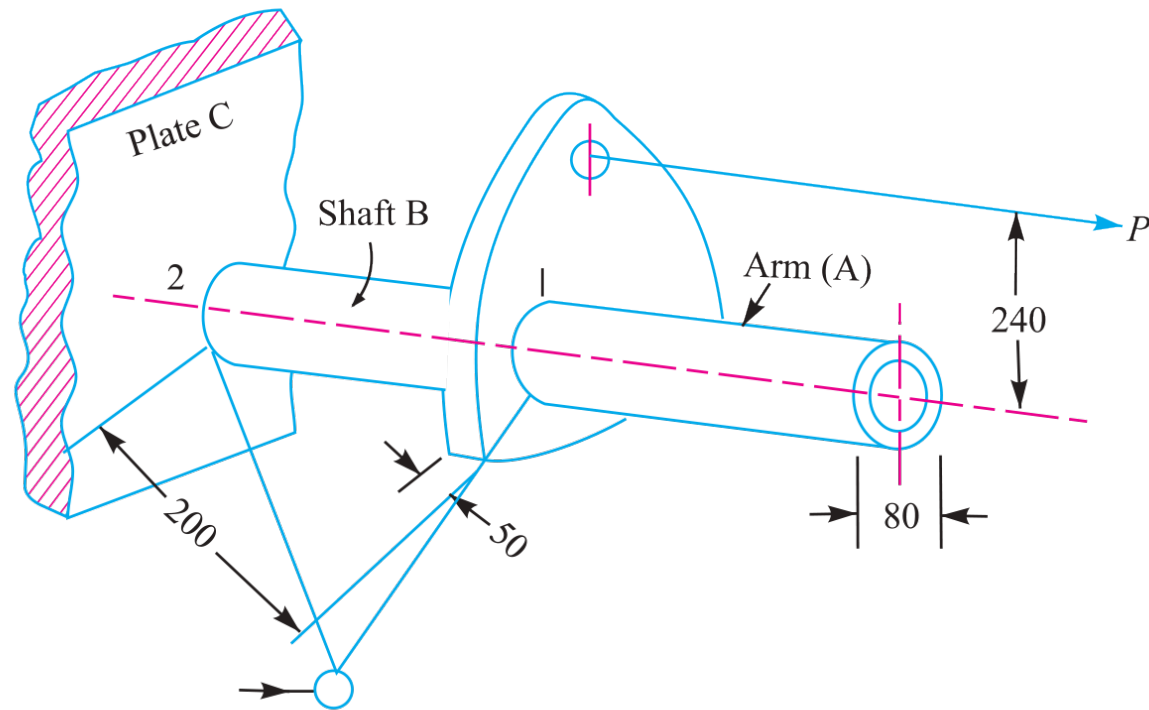


Fig. 10.27. All dimensions in mm.

Figure 10.27: Arm welded to hollow shaft subjected to combined bending and torsion

SECTION 13: ECCENTRICALLY LOADED IN-PLANE BRACKET WELD

System Parameters: Load $P = 15 \text{ kN} = 15 \times 10^3 \text{ N}$, $\tau = 80 \text{ MPa}$, $b = 80 \text{ mm}$, $l = 50 \text{ mm}$, Eccentricity $e = 125 \text{ mm}$.

Design Protocol: Determine weld size s under primary direct shear and secondary torsional shear.

1. Primary Direct Shear Stress (τ_1)

Throat area and direct shear

$$\begin{aligned} \tau_1 &= \frac{P}{2 \cdot 0.707s \cdot l} \\ &= \frac{15 \times 10^3}{1.414s \times 50} \\ &= \frac{212}{s} \text{N/mm}^2 \end{aligned}$$

2. Secondary Shear Stress (τ_2)

Polar moment of inertia and maximum radius

$$J = \frac{0.707s \cdot l(3b^2 + l^2)}{6}$$

$$J = \frac{0.707s \times 50[3(80)^2 + (50)^2]}{6} = 127850s$$

$$r_2 = \sqrt{(40)^2 + (25)^2} = 47 \text{ mm}$$

$$\cos \theta = \frac{25}{47} = 0.532$$

$$\tau_2 = \frac{P \cdot e \cdot r_2}{J}$$

$$= \frac{15 \times 10^3 \times 125 \times 47}{127850s} = \frac{689.3}{s} \text{ N/mm}^2$$

3. Resultant Shear Stress & Weld Size (s)

Vector addition of primary and secondary shear

$$\tau = \sqrt{\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta}$$

$$80 = \sqrt{\left(\frac{212}{s}\right)^2 + \left(\frac{689.3}{s}\right)^2 + 2\left(\frac{212}{s}\right)\left(\frac{689.3}{s}\right) \cos \theta}$$

$$80 = \frac{822}{s}$$

$$s = \frac{822}{80}$$

$$= 10.3 \text{ mm}$$

4. Design Output Summary
In-Plane Bracket Weld Size

Required Weld Size $s = 10.3 \text{ mm}$

SECTION 13: ECCENTRICALLY LOADED IN-PLANE BRACKET WELD — MECHANICAL
DIAGRAM

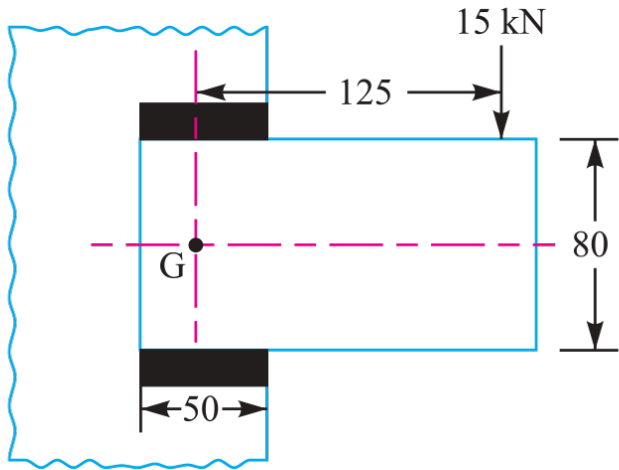


Fig. 10.28

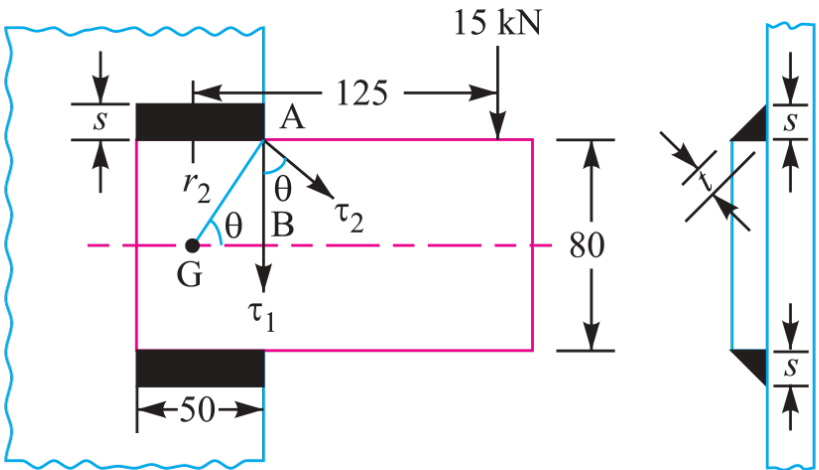


Fig. 10.29

All dimensions in mm.

Figure 10.28 & 10.29: Bracket carrying eccentric load and polar centroid vector diagram

SECTION 14: CANTILEVER PLATE CHANNEL WELD UNDER ECCENTRIC LOADING

System Parameters: Load $P = 60 \text{ kN} = 60 \times 10^3 \text{ N}$, $b = 100 \text{ mm}$, $l = 50 \text{ mm}$, Allowable Shear Stress $\tau = 140 \text{ MPa}$.

Design Protocol: Determine weld size s for channel section weld group under eccentric load.

1. Centroid (G) & Polar Moment of Inertia (J)

Centroid distance x and polar moment

$$\begin{aligned} x &= \frac{l^2}{2l + b} \\ &= \frac{(50)^2}{2(50) + 100} = 12.5 \text{ mm} \\ J &= 0.707s \left[\frac{(100 + 100)^3}{12} - \frac{(50)^2(150)^2}{200} \right] \\ &= 275 \times 10^3 \text{ smm}^4 \end{aligned}$$

2. Eccentricity (e) & Critical Radius (r_2)

Distance to load and critical corner radius

$$\begin{aligned} e &= 150 + 50 - 12.5 = 187.5 \text{ mm} \\ r_2 &= \sqrt{(50)^2 + (37.5)^2} = 62.5 \text{ mm} \\ \cos \theta &= \frac{37.5}{62.5} = 0.6 \end{aligned}$$

3. Primary & Secondary Shear Stresses (τ_1, τ_2)

Direct and torsional shear components

$$\tau_1 = \frac{P}{A}, \quad \tau_2 = \frac{P \cdot e \cdot r_2}{J}$$

$$\tau_1 = \frac{60 \times 10^3}{141.4s} = \frac{424}{s} \text{ N/mm}^2$$

$$\tau_2 = \frac{60 \times 10^3 \times 187.5 \times 62.5}{275 \times 10^3 s} = \frac{2557}{s}$$

4. Resultant Shear Stress & Weld Size (s)

Vector combination of shear stresses

$$\tau = \sqrt{\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta}$$

$$140 = \sqrt{\left(\frac{424}{s}\right)^2 + \left(\frac{2557}{s}\right)^2 + 2\left(\frac{424}{s}\right)\left(\frac{2557}{s}\right)\cos \theta}$$

$$140 = \frac{2832}{s}$$

$$s = \frac{2832}{140}$$

$$= \mathbf{20.23 \text{ mm}}$$

5. Design Output Summary

Channel Cantilever Weld Size

Required Weld Size $s = 20.23 \text{ mm}$

SECTION 14: CANTILEVER PLATE CHANNEL WELD — MECHANICAL DIAGRAM

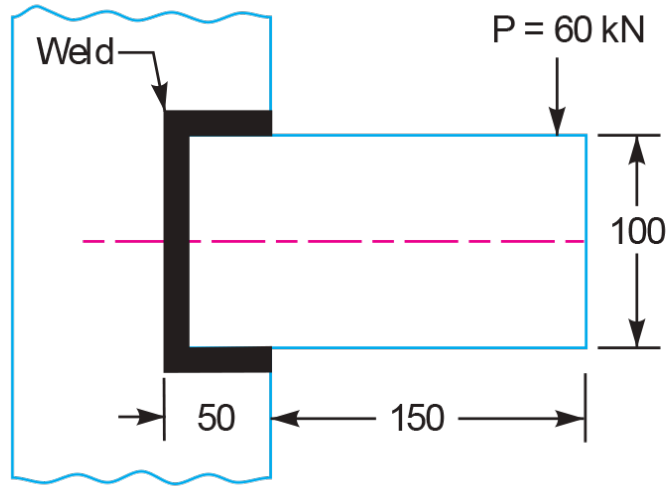


Fig. 10.30

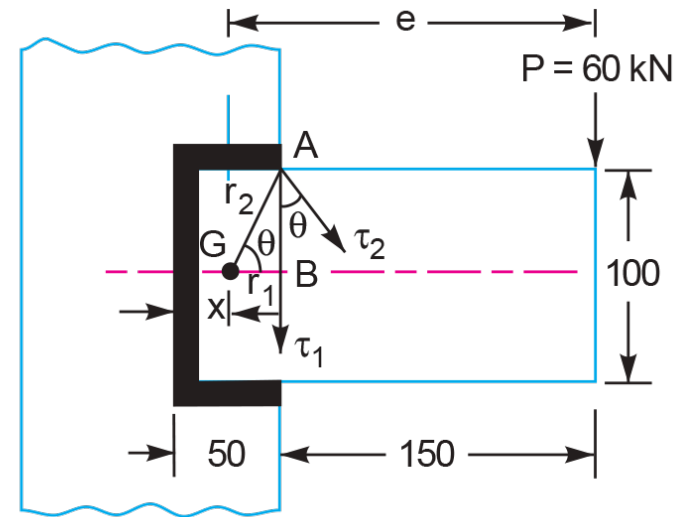


Fig. 10.31

All dimensions in mm.

Figure 10.30 & 10.31: Cantilever plate welded with channel weld pattern and centroid G

SECTION 15: CHANNEL SECTION WELDED TO PLATE UNDER ECCENTRIC IN-PLANE LOADING

System Parameters: Weld Size $s = 6 \text{ mm}$, Load $P = 20 \text{ kN} = 20 \times 10^3 \text{ N}$, $l = 40 \text{ mm}$, $b = 90 \text{ mm}$, Distance to load = 200 mm .

Design Protocol: Determine maximum shear stress $\tau_{\max}$ induced in 6 mm weld.

1. Centroid (G) & Polar Moment of Inertia (J)

Location of C.G. and polar moment of throat

$$\begin{aligned} x &= \frac{l^2}{2l + b} \\ &= \frac{(40)^2}{2(40) + 90} = 9.4 \text{ mm} \\ J &= 0.707(6) \left[\frac{(90 + 80)^3}{12} - \frac{(40)^2(130)^2}{170} \right] \\ &= 1062.2 \times 10^3 \text{ mm}^4 \end{aligned}$$

2. Eccentricity (e) & Critical Radius (r_2)

Load arm and radius to farthest weld point

$$\begin{aligned} e &= 200 - 9.4 = 190.6 \text{ mm} \\ r_2 &= \sqrt{(45)^2 + (30.6)^2} = 54.4 \text{ mm} \\ \cos \theta &= \frac{30.6}{54.4} = 0.5625 \end{aligned}$$

3. Primary & Secondary Shear Stresses (τ_1, τ_2)

Direct and torsional shear components

$$\tau_1 = \frac{P}{A}, \quad \tau_2 = \frac{P \cdot e \cdot r_2}{J}$$

$$\tau_1 = \frac{20 \times 10^3}{721.14} = 27.7 \text{ N/mm}^2$$

$$\tau_2 = \frac{20 \times 10^3 \times 190.6 \times 54.4}{1062.2 \times 10^3} = 195.2$$

4. Maximum Resultant Shear Stress (τ_{max})

Resultant shear stress in 6 mm weld

$$\tau_{\text{max}} = \sqrt{\tau_1^2 + \tau_2^2 + 2\tau_1\tau_2 \cos \theta}$$

$$= \sqrt{(27.7)^2 + (195.2)^2 + 2 \times 27.7 \times 195.2 \times \cos 90^\circ}$$

$$= \sqrt{44903.6}$$

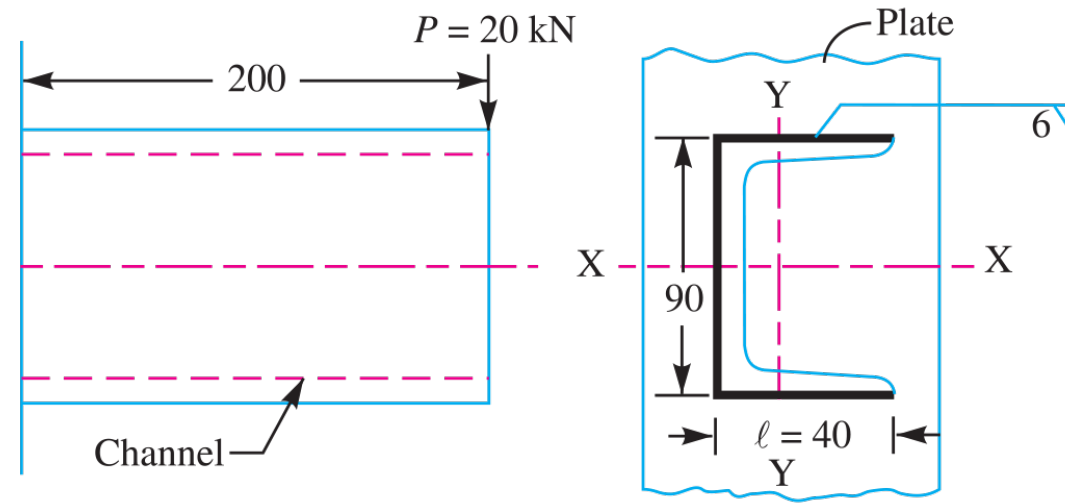
$$= \mathbf{212 \text{ MPa}}$$

5. Design Output Summary

Channel Weld Maximum Stress

Maximum Shear Stress tau_max = 212 MPa

SECTION 15: CHANNEL SECTION WELDED TO PLATE — MECHANICAL DIAGRAM



All dimensions in mm.

Fig. 10.32

Figure 10.32: Channel section welded to plate and loaded at 200 mm distance

SECTION 15: CHANNEL WELD GROUP CENTROID — MECHANICAL DIAGRAM

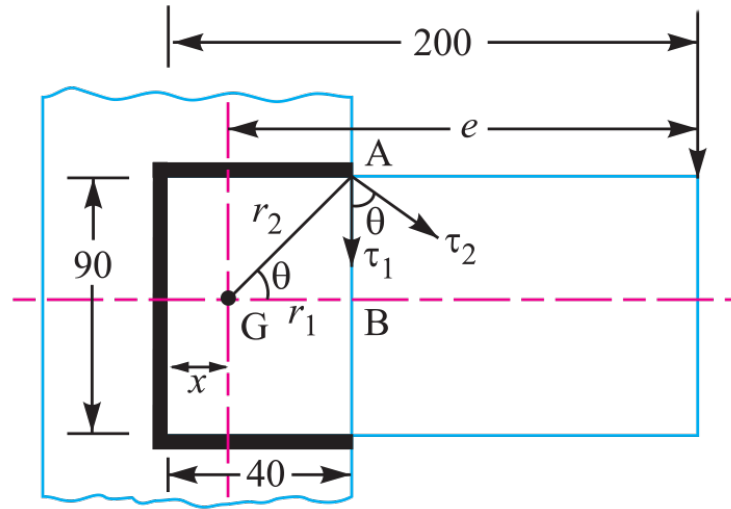


Fig. 10.33

Figure 10.33: Location of centroid G and critical radius r_2

SECTION 16: SHIELDED ARC WELDED BRACKET UNDER DEAD WEIGHT

System Parameters: Dead Weight $P = 15 \text{ kN} = 15 \times 10^3 \text{ N}$, Permissible Strength $\tau = 150 \text{ MPa}$, Weld Length $l = 25 \text{ mm}$.

Design Protocol: Calculate top (s_1) and bottom (s_2) fillet weld sizes.

1. Vertical Forces at A and B (P_{VA}, P_{VB})

Equal vertical load sharing

$$\begin{aligned} P_{VA} = P_{VB} &= \frac{P}{2} \\ &= \frac{15}{2} = 7.5 \text{ kN} \\ &= 7500 \text{ N} \end{aligned}$$

2. Horizontal Force at Top A (P_{HA})

Moment equilibrium about bottom point B

$$\begin{aligned} P_{HA} \times 75 &= 15 \times 50 \\ P_{HA} &= \frac{750}{75} \\ &= 10 \text{ kN} = 10000 \text{ N} \end{aligned}$$

3. Size of Top Fillet Weld (s_1)

Resultant force and top weld capacity

$$P_A = \sqrt{(P_{VA})^2 + (P_{HA})^2}$$

$$P_A = \sqrt{(7.5)^2 + (10)^2} = 12.5 \text{ kN} = 12500 \text{ N}$$

$$12500 = 0.707s_1 \times 25 \times 150$$

$$12500 = 2650s_1$$

$$s_1 = \frac{12500}{2650}$$

$$= 4.7 \text{ mm}$$

4. Size of Bottom Fillet Weld (s_2)

Bottom weld designed for vertical force only

$$P_{VB} = 0.707s_2 \cdot l \cdot \tau$$

$$7500 = 0.707s_2 \times 25 \times 150$$

$$7500 = 2650s_2$$

$$s_2 = \frac{7500}{2650}$$

$$= 2.83 \text{ mm}$$

5. Design Output Summary

Shielded Arc Bracket Weld Sizes

Top Weld Size $s_1 = 4.7 \text{ mm}$, Bottom Weld Size $s_2 = 2.83 \text{ mm}$

SECTION 16: SHIELDED ARC WELDED BRACKET — MECHANICAL DIAGRAM

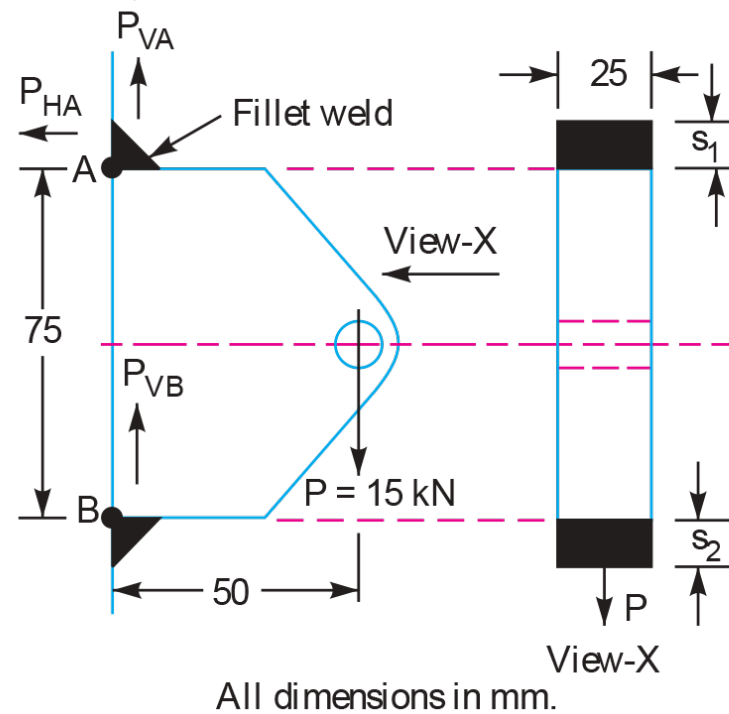


Fig. 10.34

Figure 10.34: Bracket supporting 15 kN dead weight with top and bottom fillet welds